

# LOIS S. WONG

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## EDUCATION

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### Johns Hopkins University

*Master of Science in Engineering in Computer Science*

GPA: 3.75; Awards and Honors: Rubenstein Fellowship

Master Thesis: GAITA: a RAG System for Personalized Computer Science Education (advised by Dr David Yarowsky)

### University of California, Berkeley

*Professional Certificate in Machine Learning and Artificial Intelligence*

*Bachelor of Arts in Linguistics, Minor in English*

GPA: 3.90; Honors and Societies: Phi Beta Kappa, Distinction in General Scholarship

### West Valley College

*Associate of Science in Administration of Justice*

GPA: 3.89; began coursework at age 15 following a homeschool education

## TALKS AND WORKSHOPS

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**AI in Academic Work: Capabilities, Limitations, and Responsible Us** | Invited Speaker, University of Chicago Law School, May 11, 2026

**AI in Academic Librarianship: What This Work Looks Like in Practice** | Guest Lecture, University of South Carolina, Apr 16, 2026

**What AI Can and Cannot Do** | Workshop, Grad @ UChicago, Feb 17, 2026

**Pulling the Wool Over AI** | Guest Lecture, UChicago's HUMA 10001 Undergraduate Research: What, Why, and How, Feb 10, 2026

**Using AI in Research and Writing** | Workshop, UChicago Center for Digital Scholarship, Feb 5, 2026

**Pulling the Wool Over AI** | Workshop, UChicago Center for Digital Scholarship, Nov 18, 2025

**Responsible AI Use in Humanities Research** | Guest Lecture, UChicago's GNSE Thesis Seminar, Oct 24, 2025

**GAITA: A Rag System for Personalized Computer Science Education** | Research Talk, Apple, Sep 18, 2024

## PAPERS

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Wong, L. (2024). Gaita: A RAG System for Personalized Computer Science Education. [doi.org/10.35542/osf.io/97nmg](https://doi.org/10.35542/osf.io/97nmg)

Wong, Lois Sun, Ehong. (2024). Balancing Personalization, Privacy, and Ethical Challenges in Recommender Systems. 10.13140/RG.2.2.26142.40007.

Wong, L., Ferber, A. (2024, July 11). Advancing Personalized Computer Science Education: An Information Retrieval Perspective. [doi.org/10.35542/osf.io/sndum](https://doi.org/10.35542/osf.io/sndum)

Wong, L. (2023). On the Use of Metaphor Translation in Psychiatry. *arXiv preprint arXiv:2312.14845*.

## PROFESSIONAL EXPERIENCE

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### University of Chicago, Center for Digital Scholarship

Sep 2025 – Present

*AI Librarian*

- Teaching workshops and guest lectures on Introductory AI and Ethical/Responsible usage in research and learning
- Developing risk evaluation frameworks for evaluating AI projects and vendor proposals to ensure responsible AI integration
- Providing research consultations to faculty and students
- Developing internal AI education resources for staff (Canvas course, resource guides, monthly newsletter)
- PM for the library's strategic priority #11

## **Novita AI**

*June 2025 – Sep 2025*

### *Founding Developer Relations Engineer*

- Established scalable systems for email campaigns, automations, and CRM use to drive top-of-funnel growth
- Built developer relations and content strategy, including user onboarding flow, product education, and user-facing resources for AI infrastructure products (nurture leads)
- Created documentation, blog posts, and marketing/educational materials that helped developers understand and integrate Novita's tools into their workflows
- Represent Novita through event presence (presenting hackathon awards, tabling at AI events)

## **AIML.com**

*Jan 2025 – May 2025*

### *Machine Learning Content Writer*

- Write SEO-optimized articles and interactive case studies to explain complex ML and AI concepts for technical and non-technical audiences (published on AIML.com)
- Collaborate with editors to refine content, ensuring clarity, technical accuracy, and accessibility

## **Johns Hopkins Center for Digital Health and AI**

*Jan 2025 – Mar 2025*

### *Technical Lead*

- Lead development of a RAG chatbot enabling interactive engagement with AIM-AHEAD's Data Science coursework
- Write technical design documentation to outline rationale and implementation details for stakeholders
- Present project updates and outcomes to stakeholders

## **Apple**

*Aug 2023 – Sep 2024*

### *AI Education Teaching Assistant (AIML org)*

- Collaborated with Industry Experts Educators to develop and produce a series of internal NLP and ML courses
- Designed and implemented hands-on coding exercises using Jupyter notebooks to provide practical experience
- Established and adhered to style guides to maintain consistency in course materials
- Delivered a research talk on Retrieval-Augmented Generation (RAG) and its educational applications

## **International Rescue Committee**

*June 2024 – Aug 2024*

### *Data Analytics and Systems Intern*

- Designed and implemented a SharePoint-based Knowledge Management System with an integrated RAG chatbot to streamline information access and enhance operational efficiency
- Conducted user research to identify user needs, optimizing data retrieval workflows for refugee aid programs

## **Grade Potential Tutoring**

*Feb – Aug 2023*

### *Academic Tutor*

- Created customized learning tactics and strategies for K-12 students to enhance their proficiency in English and Math
- Motivated and supported struggling students to regain confidence and achieve significant grade improvements
- Edited and proofread students' papers, offering corrections and suggestions to enhance grammar, clarity, and flow

## **SELECTED TECHNICAL PROJECTS**

### **GAITA: Personalized Pathways for Learning Computer Science**

*Apr 2024 – Dec 2024*

- Developed and deployed GAITA, a GPT-4o-mini RAG chatbot that creates personalized CS learning pathways tailored to users' background and goals
- Built a web crawler & scraper to compile a vector-enhanced database of 1,200+ open-access CS courses
- Created prompt templates and system architecture for reliability and accuracy
- Implemented a semantic retrieval pipeline for course recommendations
- Built learning pathways through iterative prompting
- Selected for the Backdrop Build pre-accelerator and presented at a Research Talk at Apple

### **Deep Learning Course Development for Apple Engineers**

*Mar 2024 – Sep 2024*

- Designed and implemented hands-on coding exercises using Jupyter notebooks to reinforce technical concepts
- Received positive feedback from pilot participants, highlighting the course's relevance and effectiveness
- Established and adhered to style guides to maintain consistency in course materials, ensuring alignment with Apple's internal education standards

### **Mitigating Social Bias in Language Models Through Adversarial Debate**

*Apr – May 2024*

- Designed and implemented an experimental framework using GPT-3.5 turbo to explore and measure biases by promoting debates on stereotypes

- Analyzed the effectiveness of debate interventions on pre- and post- debate model outputs
- Implemented a standardized question framework to measure shifts in model viewpoints and assess bias reduction
- Developed metrics and reports to gain insights into bias mitigation

### **Suicide Ideation Detection**

*Jul – Sep 2022*

- Preprocessed over 20,000 Reddit posts using NLP techniques to prepare data for classification
- Trained text classification models achieving 92% accuracy and 96% recall in detecting suicide ideation
- Utilised ELI5 and LIME for global and local model interpretability, validated feature importances using KL-Divergence scores, and developed a vocabulary of predictive terms
- Received recognition from professors and was selected as an exemplar project for future cohorts

### **Research Study Replication**

*Dec 2021*

- Replicated "Expressive Intent, Ambiguity, and Aesthetic Experiences of Music and Poetry" using raw data in R
- Applied Linear Mixed Effects models to investigate how knowledge of authorial intent impacts aesthetic experience
- Conducted statistical analyses (ANOVA, t-tests, F-tests) to evaluate the results and address the research question

## **RESEARCH EXPERIENCE**

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### **University of Michigan**

*Sep 2022 – Mar 2023*

#### *NLP Research Intern*

- Analyzed and synthesized cutting-edge NLP research, transforming key concepts into a structured knowledge graph
- Added over 180 contributions to the collaborative research platform, regularly presenting updates and key findings

### **San Jose State University**

*Mar – Aug 2022*

#### *Research Assistant to Professor Jonathan Rawski—Computational Phonology*

- Analyzed computational analyses in linguistics, focusing on Automata, Grammars, and Formal Language Theory
- Investigated the impact of various formal grammars on encoding phonological structures and processes

## **SELECTED LINGUISTICS RESEARCH PROJECTS**

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### **Semantic Comparison of French and English Definite Articles**

- Applied Lambda Calculus to model implication, presupposition, and degrees of specificity across French and English
- Collected and annotated spoken data in both languages, adhering to rigorous linguistic standards
- Applied interlinear glossing to demarcate syntactic categories and modelled syntactic structure with trees
- Addressed natural language ambiguity by converting text data into First Order Logic

### **Syntactic Analysis of Seenku**

- Proposed an X-Bar analysis of Seenku by reviewing 150+ documents of corpus data
- Derived a phrase structure grammar from observed syntactic patterns, and modelled syntactic structure with trees
- Developed recursive generative rules to analyze question formation, antipassivation, and DP movement

### **Phonological and Typological Analysis of Seenku**

- Compiled data from a grammar to analyse Seenku's phonological processes of palatalization and affrication
- Established phonological rules to describe the transformations involved in these processes
- Summarized Seenku's phoneme inventory, syllable structure, morphology, and tone systems, situating the language within a typological perspective

### **Phonetic Analysis of Mandarin**

- Collected and phonetically transcribed 120+ instances of spoken Mandarin data using IPA
- Annotated spectrograms with Praat to illustrate phonetic segmentation and demarcate tonal and vowel alternation
- Measured the effects of bilingualism on pronunciation, comparing meaningful differences in Mandarin and Cantonese
- Validated findings through interviews with native speakers, ensuring accuracy and relevancy of the phonetic analysis

## **AWARDS AND HONORS**

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Rubenstein Fellowship – Johns Hopkins University, 2023

2nd Place – Intel® AI for Social Good Hackathon, NLP Track, 2022

Distinction in General Scholarship – UC Berkeley, 2021

Phi Beta Kappa – UC Berkeley, 2021

Berkeley Transfer Scholarship – UC Berkeley, 2020

Distinction – Associate of Trinity College London (ATCL) in Violin Performance Diploma, 2018